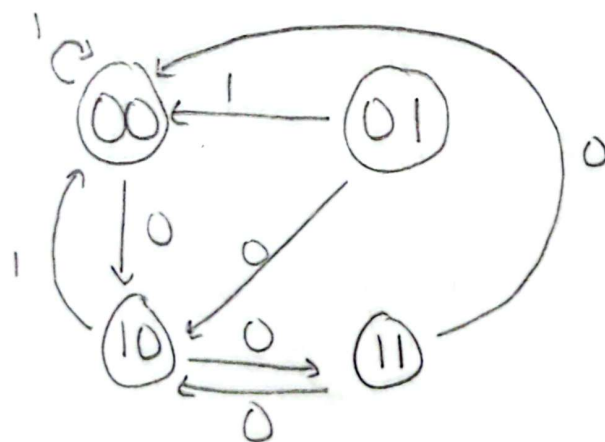


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Ans. to the ques. no. 1

A	B	x	JA	KA	DB	A ⁺	B ⁺
0	0	0	1	1	0	1	0
0	0	1	0	1	0	0	0
0	1	0	1	1	0	1	0
0	1	1	0	1	0	0	0
1	0	0	0	0	0	1	1
1	0	1	0	1	0	0	0
1	1	0	0	0	0	1	0
1	1	1	0	1	0	0	0



Ans. to the ques. no. 2

Total states = 4

$$\text{no. of flipflops} = \lceil \log_2(4) \rceil$$
$$= 2$$

Let,

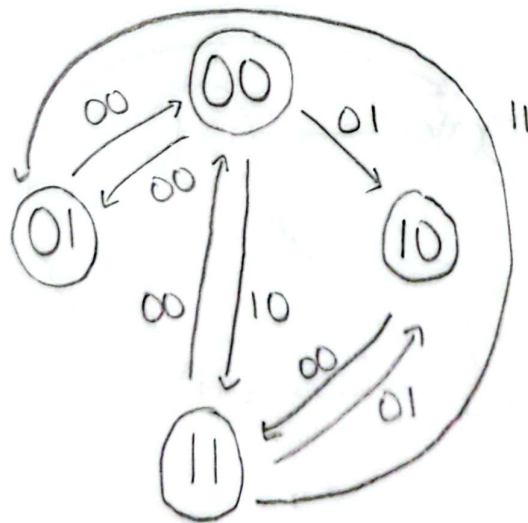
Leon = 00

Claire = 01

Ada = 10

Chris = 11

$$\text{No. of external input} = \lceil \log_2(3) \rceil$$
$$= 2$$



Dec	A	B	X	Y	A ⁺	B ⁺	TA	TB
0	0	0	0	0	0	1	0	1
1	0	0	0	1	1	0	1	0
2	0	0	1	0	1	1	1	1
3	0	0	1	1	X	X	X	X
4	0	1	0	0	0	0	0	1
5	0	1	0	1	X	X	X	X
6	0	1	1	0	X	X	X	X
7	0	1	1	1	X	X	X	X
8	1	0	0	0	1	1	0	1
9	1	0	0	1	X	X	X	X
10	1	0	1	0	X	X	X	X
11	1	0	1	1	0	1	1	1
12	1	1	0	0	0	0	1	1
13	1	1	0	1	0	0	0	1
14	1	1	1	0	X	X	X	X
15	1	1	1	1	X	X	X	X

Now,

TA

$AB \backslash xy$	$x'y'$	$x'y$	xy	xy'
$A'B'$	0	1	X	2
$A'B$	X	5	X	6
AB	12	13	X	14
AB'	8	9	11	X

$$TA = A'y + X + AB y'$$

TB

$AB \backslash xy$	$x'y'$	$x'y$	xy	xy'
$A'B'$	X	1	X	2
$A'B$	X	5	X	6
AB	12	13	X	14
AB'	8	X	11	X

$$TB = A + y'$$

